

Below is our HOME IP Address and Default Gateway:

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
IPv4 Address. : 10.0.0.177
Subnet Mask : 255.255.255.0
Default Gateway : 10.0.0.1

What IP Address and Gate Way IP Address should i use inside my Windows Servers (DC, Node1 and Node2) to be added to domain windows server hosted on a Oracle VM Virtual BOX without IP Conflict with Host Windows 10 ?

C:\Users\DELL>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 10:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
IPv4 Address. : 10.0.0.177
Subnet Mask : 255.255.255.0
Default Gateway : 10.0.0.1

C:\Users\DELL>

To avoid **IP conflicts with your host (Windows 10) network**, which uses:

- **Network:** 10.0.0.0/24
- **Host IP:** 10.0.0.177
- **Gateway:** 10.0.0.1

We must ensure our **Oracle VirtualBox VMs (DC, Node1, Node2)** are on a **different, isolated network**.

Goal Expectation

We are hosting:

- **Windows Server Domain Controller (DC)**
- **Node1**
- **Node2**

Inside **Oracle VM VirtualBox** on **Windows 10**, and want:

- VMs to **communicate with each other** (domain functionality)
- No **IP conflicts with host's Wi-Fi (10.0.0.0/24)**
- Optional: host-to-VM communication or internet (based on your use case)

 Network Mode Options in VirtualBox

Mode	Description	Internet?	Isolated from Host Network?	VM ↔ VM?
NAT	Shared IP via host	✓	✓	♦ (Limited)
Bridged	Directly on host's network	✓	✗	✓
Host-Only	Private network between VMs and host	✗	✓	✓
Internal	Fully private network (VM ↔ VM only)	✗	✓	✓

✓ **Best for Domain Lab: Host-Only Adapter** (or Internal if you don't need host access).

 **Recommended Setup: Use "Host-Only Adapter"**

✓ **Step-by-Step Plan**

1. Create a Host-Only Network in VirtualBox

- Open **Oracle VM VirtualBox**
- Go to **File > Tools > Network > Host Network Manager**
- Click **Create**
- You'll get something like:
 - **Interface Name:** vboxnet0
 - **IPv4 Address:** 192.168.56.1
 - **Subnet Mask:** 255.255.255.0
- Disable **DHCP** (you'll assign static IPs manually)

✓ This creates an **isolated network**: 192.168.56.0/24, separate from your home 10.0.0.0/24

2. Attach VMs to Host-Only Network

For **each VM** (DC, Node1, Node2):

- Go to **Settings > Network**
- **Enable Adapter 1**
 - Set to: Attached to: Host-only Adapter
 - Choose: vboxnet0

3. Assign Static IP Addresses to Each VM

Use this table:

VM	IP Address	Subnet Mask	Gateway (dummy)	DNS Server
DC	192.168.56.10	255.255.255.0	192.168.56.1	127.0.0.1 or 192.168.56.10
Node1	192.168.56.11	255.255.255.0	192.168.56.1	192.168.56.10
Node2	192.168.56.12	255.255.255.0	192.168.56.1	192.168.56.10

 The gateway here is not real — it just ensures Windows doesn't complain. No internet is needed for domain setup.

Optional: Add Second Adapter for Internet Access

If you want your VMs to have **internet access** (e.g., for Windows Updates):

1. Go to **Settings > Network**
2. **Enable Adapter 2**
 - Attach to: **NAT**

Now each VM will have:

- **Adapter 1:** Host-Only for domain traffic
- **Adapter 2:** NAT for internet

Summary

- **Avoid using 10.0.0.x** on VMs — that's your home network, will cause IP conflicts
- Use **Host-Only network (192.168.56.0/24)** for DC and domain members
- Assign **static IPs**:
 - DC: 192.168.56.10
 - Node1: 192.168.56.11
 - Node2: 192.168.56.12
- DNS should point to **DC**